University of Toronto Scarborough
Department of Computer and Mathematical Sciences

December Examinations 2015

CSC A67H3 F / MAT A67H3 F

Duration—3 hours

Aids allowed: None; closed-book, no calculators.

Make sure that your examination booklet has 13 pages (including this one). Write your answers in the spaces provided. You will be rewarded for concise, well-thought-out answers, rather than long rambling ones. Please write legibly.

This exam was designed so that you have plenty of time to write it. Take a few minutes before you begin to read each question, and then start with the question you are most comfortable with. **Since you do not have use of a calculator, you are not expected to resolve all answers to a final number. Only do so when the calculation is possible without a calculator.**

Name: ____________________________ UTORid: ________________________
(Circle your family name.)

Student #: ________________________ Tutorial section: ________________________

YOU MUST SIGN THE FOLLOWING:

I declare that this exam was written by the person whose name and student # appear above.

Signature: ____________________________

Your grade

1.		/ 10	*6.*		/ 10
2.		/ 5	*7.*		/ 10
3.		/ 15	*8.*		/ 10
4.		/ 10	*9.*		/ 10
5.		/ 15	*10.*		/ 10
			Total		/ 105

Question 1

[10 marks]

Your local grocery store just received a large shipment of apples, oranges, pears, and bananas—more than 100 pieces each. You are shopping at the store and will purchase your fruit for the week.

a. How many ways can you select 10 pieces of fruit from your store's supply of apples, oranges, pears, and bananas?

b. How many ways can you select 10 pieces of fruit from your store's supply of apples, oranges, pears, and bananas if you need at least one piece of each kind of fruit?

CONTINUED . . .

Question 2

[5 marks]

In how many ways can six 3's and four 2's be arranged in a row so that the 2's are always apart?

Question 3

[15 marks]

Your Toronto Maple Leafs won 30 of 82 games last season (i.e., the 2014-2015 season), giving them a winning percentage of 37%. If we assume this means the probability of the Leafs winning any given game is 0.37, then we can predict how they would have done in a playoff series.

Answer the following questions to determine the probability that the Leafs would have won a best of 7 playoff series (i.e., won 4 games) had they made the playoffs last season.

a. Rephrase this question in terms of sequences of 0s and 1s. What is the shortest length of a sequence? What is the longest length of a sequence?

b. Calculate the number of sequences which correspond to the Leafs winning the series. (Note that the answer is *not* $C(7, 4)$.)

c. Calculate the number of sequences as they relate to this problem. (Note that the answer is not 2^7 as not all series would last 7 games.)

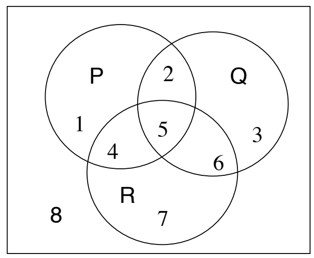
d. Calculate the probability that the Leafs would win the series.

e. What is your best guess for the probability that the Leafs will ever win the Stanley Cup again (the ultimate prize in the NHL)?

Question 4

[10 marks]

For each of the following statements, list the numbered region(s) of the venn diagram that represent(s) when the statement is true.



a. $P \wedge Q \wedge \neg R$

b. $(Q \to P) \to R$

c. $\neg((P \wedge \neg Q) \to \neg R)$

Question 5

[15 marks]

Logical equivalence and contrapositive.

a. Circle the statements below that are equivalent to $(a \wedge b) \rightarrow c$.

(1) $(\neg a \vee \neg b) \rightarrow \neg c$
(2) $\neg c \rightarrow (\neg a \vee \neg b)$
(3) c is necessary for a and b
(4) c is sufficient for a and b
(5) a and b are sufficient for c
(6) a and b are necessary for c
(7) all of the above
(8) none of the above

b. If the implication $a \rightarrow b$ is difficult to prove directly, we can attempt a *proof by contrapositive*. Use a truth table to show that the contrapositive of $a \rightarrow b$ is equivalent to $a \rightarrow b$.

c. Use logical equivalences to prove that the contrapositive of $a \to b$ is equivalent to $a \to b$.

Logical Equivalences

$p \to q \Leftrightarrow \neg p \vee q$	Arrow Law
$p \wedge q \Leftrightarrow q \wedge p$	Commutative Laws
$p \vee q \Leftrightarrow q \vee p$	
$p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$	Associative Laws
$p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r$	
$\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$	DeMorgan's Laws
$\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$	

Question 6

[10 marks]

Modular arithmetic.

a. State the *Division Theorem*. (**Hint**: This is the theorem we often use when writing proofs involving the **mod** operator.)

b. Prove that $\forall a \in \mathbb{N}, \forall b \in \mathbb{N}, \forall c \in \mathbb{N}, a \equiv_n b \to ca \equiv_n cb$.

Question 7

[10 marks]

Consider the following statement S:

The negative of any irrational number is irrational.

a. Write S as an *implication* I.

I:

b. Prove I.

Question 8

[10 marks]

Prime numbers and the *Fundamental Theorem of Arithmetic (FTA)*.

a. State the *FTA*.

b. Prove or disprove that for all *prime* $p \in \mathbb{N}$, $\sqrt{p}$ is irrational. (Recall that a prime number is a natural number *greater than* 1 that has no positive divisors other than 1 and itself.)

Question 9

[10 marks]

We wish to prove that for all natural numbers $n \geq 0$, $10^n - 1$ is an integer multiple of 9. Fill in the missing steps of the proof:

Define $S(n)$:

Prove $S(n)$ for all $n \in \mathbb{N}$.

Base Case.

Induction Hypothesis. Assume for ______________________________ $S(k)$ is true.

Induction Step. Prove $S(k) \rightarrow S(k+1)$.

Question 10

[10 marks]

Define the function $a(n)$ as follows:

$$a(n) = \begin{cases} 5 & n = 1 \\ 10 & n = 2 \\ 2a(n-1) + a(n-2) & n \geq 3 \end{cases}$$

and let $S(n)$ represent $a(n) < 3^n$. Use either *simple* or *strong* induction to prove that $S(n)$ is true for all $n \geq 3, n \in \mathbb{N}$. (**Hint**: It is much easier to prove $S(n)$ if you choose the right form of induction!)

END OF EXAM